Phonological Degrees of Labiality: Evidence from Karata

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INTRODUCTION

Karata (*k̄'irīi macī'i*, Russian *karatinskij jazyk*): Nakh-Daghestanian family, ca. 10,000 speakers, 8 villages, North-West Daghestan, Russia.

BACKGROUND INFORMATION

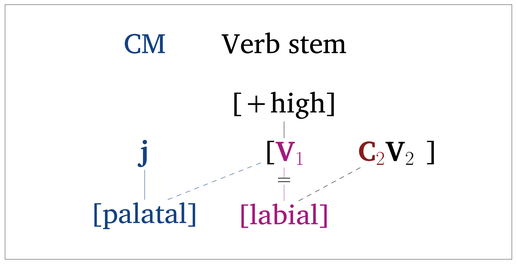
- 86 phonemic C (46 non-labialized C & 40 labialized C)
- 5 V (with long x nasal counterparts)
- Phonotactic constraints:
  1. no [round V + labialized C] sequence (***uC**^{w})
  2. no consecutive labialized onsets (***C**$^{w}_{1}C^{w}_{2}$)
- The UR of the verb stem is apparent with the singular Neutral Class Marker *b-* (This is the only CM that does not neutralize the following V contrast).
- Underlyingly non-labialized C's undergo a process of labialization

CONSONANT LABIALIZATION

- In Karata verbs, a stem C must be labialized when both following conditions are met:
  1. the preceding V is underlyingly round and high

     ✓ b-utʃ̃aɬa — N-wash b-oχ̃aɬa — N-appear
  2. this V becomes unround as a result of assimilation to a preceding CM: Fem.CM *j-*, H.pl.CM *b(a)-* or nH.pl.CM *r(a)-*

     j-itʃ^{w} ãɬa — F-wash j-eχ̃aɬa — F-appear



high	i	u
mid	e	o
low	a	

(1) high/round V_1 → C labialization
- a. /j + utʃ̃aɬa/ → jitʃ^{w}ãŋɬa `wash'
- b. /j + utʃ'ãɬa/ → jitʃ^{w}'ãŋɬa `open'
- c. /j + utaɬa/ → jit^{w}aɬa `untie'
- d. /j + uɪ̯'ãɬa/ → jiɪ̯^{w}'ãŋɬa `share'

(2) No labialization
- a. /j + oχ̃aːɬa/ → jeχ̃aːɬa `thrust'
- b. /j + oq̃aɬa/ → jeq̃aɬa `remove'
- c. /j + oχ̃aɬa/ → jeχ̃aɬa `appear'
- d. /j + oɪ̯'aɬa/ → jeɪ̯'aɬa `warm up'

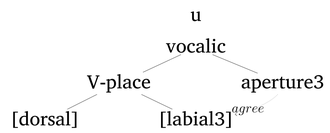
- C̄ = fortis consonant
- fortis = distinctive feature used across all art. places
- phonetic realization contingent on the type of C that realizes it

ASSUMPTIONS

VOWEL APERTURE

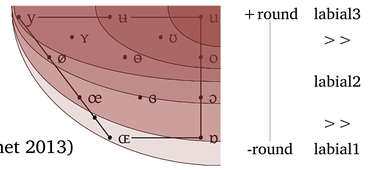
- V height is defined in terms of degrees of aperture (Clements and Hume 1995)
- `Aperture' refers to the degree to which the oral cavity is open/close

aperture 3	i	u
aperture 2	e	o
aperture 1	a	



LABIALITY DEGREES AND SCALE

- There are degrees of rounding too.
- In this language (and others), if a V has a [labial] feature, it agrees in degree with the aperture node of the same V.
- Height and backness = rounding enhancement



(Lionnet 2013)

- The roundedness scale maps vowel height to labiality degree.
- `The faithfulness constraints are naturally ranked so that a more prominent member of the scale demands more faithfulness.' (Gnanadesikan 1997)

EVIDENCE FOR LABIALITY DEGREES

- **Articulatory evidence (Linker 1982)**: the rounding gesture of a high vowel is more extreme than the rounding gesture of a non-high vowel.
- **Perceptual evidence (Terbeek 1977)**: high rounded vowels are perceived as more rounded than non-high rounded vowels.
- **Typology (Maddieson 1984)**: the lower the rounded vowel, the rarer it is.
- **Contrastive labiality gestures**:
  1. Swedish: 2 high front round V [yː, ʉ], [ʉ] is not as protruded (Linker 1982)
  2. Assamese: 2 low back round V [ɑ,ɒ] (Ladefoged et al. 1996)

SKETCH OF THE ANALYSIS

- When [labial3] and [labial2] are delinked as a result of assimilation, there is markedness pressure for delinked features to delete.
- But there is more pressure to keep [labial3] than the other [labial *x*] features.
- It is better to re-associate [labial3] to another segment than to delete it.

ANALYSIS

CONSTRAINTS

- **Anti-mismatch constraint**

  *$\overset{\text{aperture}_{\alpha}}{\underset{\text{labial}_{\neg\alpha}}{\text{x}}}$ — Assign a violation for every output rootnode which is linked to an aperture node and a labial feature which disagree in value.
- **Assimilation constraint**

  * $\overset{\text{O}}{\text{V}_{\alpha}}$ $\overset{\text{N}}{\text{V}_{\neg\alpha}}$ — Assign a violation for every output onset-nucleus sequence if their V-place nodes are not the same.
- ***FLOAT** — Assign a violation for every feature that is floating in the output (Wolf 2007).
- **MAX [labial3] > > MAX [labial2]**

  MAX[lab*x*] — Assign a violation for every [labial*x*] feature that is not present in the output.
- **Anti-labialization constraint**

  IdentOI[lab*x*] — Assign a violation for every link between a [labial*x*] feature and a segment in the output that is not in the input.

ANALYSIS (SIMPLIFIED)

/j ~~u~~č / [lab3]	*O$_{\alpha}$N$_{\neg\alpha\,PLACE}$	Max[labial3]	*Float	*ap$_{\alpha}$/labial$_{\neg\alpha}$	IdentIO[labial*x*]	Max[labial2]
a j ~~u~~č^{w} [lab3]					*	
b j i č [lab3]			*W		L	
c j i č ~~[lab3]~~		*W			L	
d j u č [lab3]	*W				L	

/j ~~o~~č / [lab2]	*O$_{\alpha}$N$_{\neg\alpha\,PLACE}$	Max[labial3]	*Float	*ap$_{\alpha}$/labial$_{\neg\alpha}$	IdentIO[labial*x*]	Max[labial2]
a j e^{w} č [lab2]					*W	L
b j e č [lab2]			*W			L
c j e č ~~[lab2]~~						*
d j o č [lab2]	*W					L

ANTICIPATING QUESTIONS

- Phonology can access gradient subphonemic information
- Is this information phonetic or phonological?

Claim: it is phonological.
Karata and languages with parasitic rounding harmony have phonologized phonetic reflexes.

- The degree of rounding is predictable from V height: is it necessary to have degrees of rounding on [labial]?
  1. On the surface, rounding is **not** on V anymore so we can't refer to V height
  2. If at an intermediate level [labial] is linked to both V C, then it's a case of opaque spreading
- Why not just stipulate that [+high] V trigger opaque spreading?
  1. At an intermediate level of the derivation, we would need /uC^{w}/ to win
  2. But this won't happen because ***uC**^{w} is undominated

(*see* Pasquereau 2013 for more details)

PREDICTIONS

ROUNDING HARMONY

- Parasitic rounding harmony

Anti-mismatch > > Extend(labial*x*)

- Yawelmani Yokuts (Yokutsan): high-high low-low
- Kachin Khakass (Turkic): high-high

labial degrees are contrastive

- Cross-height rounding harmony

Extend(labial*x*) > > Anti-mismatch

- Turkish: low/high-high
- Yakut (Turkic): low-low/high high-high

labial degrees are not contrastive

(data and EXTEND from Kaun 1997)

VOWEL COALESCENCE

- In Yoruba (Niger-Congo), under coalescence, the rounding of a high vowel is preserved but not that of a mid-vowel: /u + i/ → [u] BUT /o + e/ → [e] (Casali 1996)

APERTURE/LABIAL MISMATCHES

- In languages where *ap$_{\alpha}$/lab$_{\neg\alpha}$ is low ranked, mismatches are allowed (e.g. cross-height rounding harmony)
- Is the lip gesture different from a round vowel with matching *aperture* and *labiality* degrees?

CONCLUSION

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- Karata consonant labialization
  1. [labial*x*] features agree in value with V aperture
  2. there is more faithfulness to keep [labial3] in the output
- Karata has phonologized distinctions that are traditionally considered purely phonetic
- This helps understand phenomena of parasitic rounding harmony
- Future research:
  1. Is the labial gesture on a V whose rounding results from harmony different from the gesture on a V which is underlyingly round?
  2. Do processes of V coalescence bear out the fixed-ranking of faithfulness to [labial*x*]?

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